ECE 6010
Kalman Filtering
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1. write a function that implements the Kalman filtering
2. test your function using the system in 12
3. try different values and comment on how well it is able to estimate the state.

   The Kalman filter is able to pretty accurately track the state of the position of the system. However, it does a poor job of tracking the velocity of the system. This makes sense as we don't have observations of the velocity. The following plots have different covariances and different iterations.

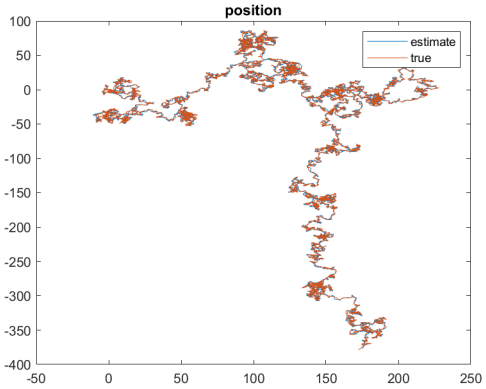


Figure 1: plot of position with estimated and true positions for test 1

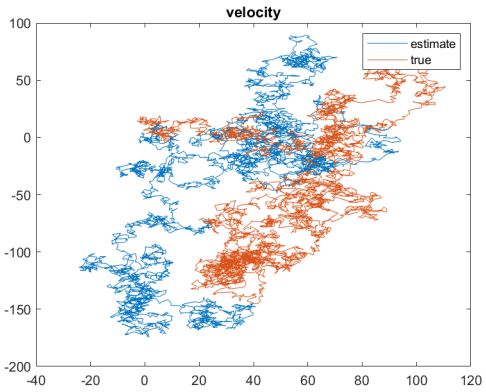


Figure 2: plot of velocity with estimated and true positions for test 1

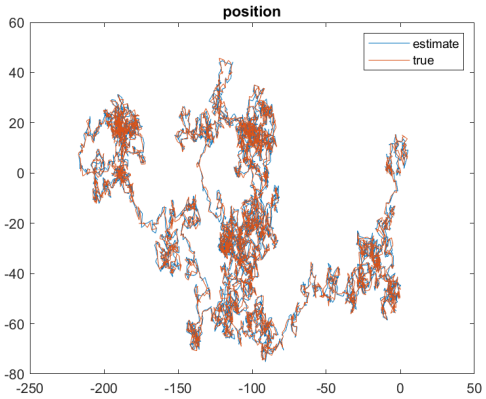


Figure 3: plot of position with estimated and true positions for test 2

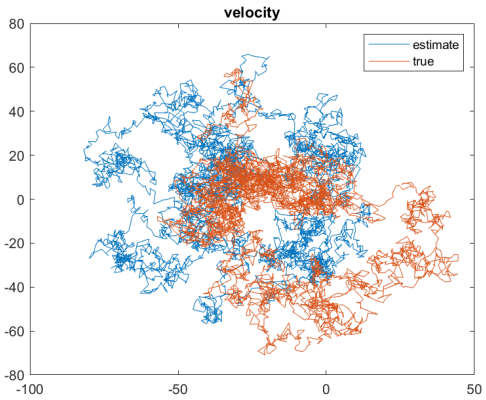


Figure 4: plot of velocity with estimated and true positions for test 2

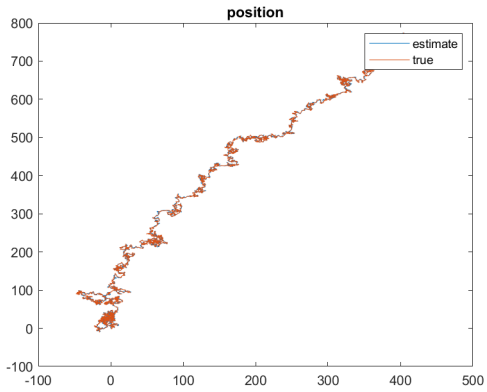


Figure 5: plot of position with estimated and true positions for test 3

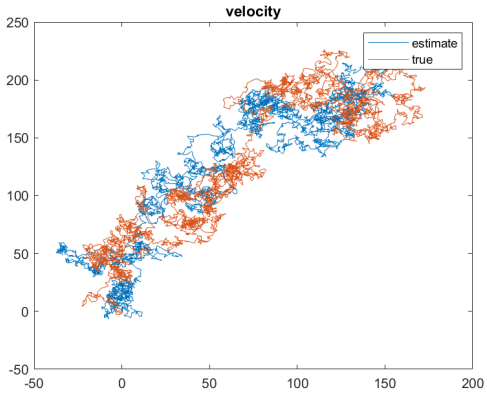


Figure 6: plot of velocity with estimated and true positions for test 3

```
function [xhat, P] = kalman(y,xtt,Ptt, A,C,Q,R)
Kt = Ptt*C.'/(C * Ptt * C.' + R);
xhatt = xtt + Kt*(y-C*xtt);
Pt = (eye(size(Kt*C)) - Kt*C)*Ptt;
xhat = A*xhatt;
P = A*Pt*A.' + Q;
end
```

Listing 1: MATLAB Code

```
clear all; clc; close all;format compact

h = 0.001;
m = 1;
Atild = [0,1;0,0];
btild = [0;1/m];
xt = zeros(2,1);
xtt = zeros(2,1);

A12 = eye(4);A12(1,2) = h;A12(3,4)=h;
B = zeros(4,2);B(2,1)=h/m;B(4,2)=h/m;
```

```
% w = B*f, f=input
w = eye(4);w(1,1)=2;w(1,2)=1;w(2,1)=1;w(2,2)=2;w(3,3)=3;w(3,4)=2;w
    (4,3)=2;
w(4,4)=3;
v = eye(2);

% simulation

xt = [0;0;0;0];P = w;xhat = xt;
Lw = chol(w,'lower');
Lv = chol(v,'lower');
ymat = [1,0,0,0;0,0,1,0];
xtrue = [];xest = [];
for i = 1:5000
    wt = randn(4,1);
    xt = A12*xt + (Lw*wt);
    vt = randn(2,1);
    yt = ymat*xt + (Lv*vt);
    [xhat,P] = kalman(yt,xhat,P,A12,ymat,w,v);
    xtrue=[xtrue,xt];xest=[xest,xhat];
end
sprintf("finished\n")
display(xhat);
display(xt)

figure(1);
plot(xest(1,:),xest(3,:));
hold on;
plot(xtrue(1,:),xtrue(3,:));
hold off;
legend('estimate', 'true');
title('position');

figure(2);
plot(xest(2,:),xest(4,:));
hold on;
plot(xtrue(2,:),xtrue(4,:));
hold off;
legend('estimate', 'true');
title('velocity');
```

Listing 2: MATLAB Code